

11.1

EEE 308 Introduction to Control Systems Test Time: 45 Min

- A feedback control system has an open-loop transfer function $G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)}$.
 (i) At what value of K will the system becomes unstable (ii) determine the value of K at damping factor 0.5.
- The open loop transfer function of a unity feedback system is $G(s) = \frac{K}{s(s+2)}$. The system is required to meet the following specifications: peak time, $t_p = 1.0$ sec. peak overshoot, $m_p = 10\%$.
 (i) Find a value of k to meet both specifications. If not, what should be the value of k so that t_p and m_p are violated by the same percentage? (ii) Assuming that the open loop pole $s = -2$ can be adjusted to $s = -\alpha$. Determine the values of k and α in order to meet both the specifications.

$$t_r = \frac{\pi - \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}}{\omega_n \sqrt{1-\zeta^2}}$$

$$M_p = e^{\frac{-\pi \zeta}{\sqrt{1-\zeta^2}}}$$

$$\frac{1}{\omega^2 s^2 + 2\omega_n s + \omega_n^2}$$

$$40.8 - 5K + 6$$

$$\frac{46.8 - 5K}{5}$$

$$t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$



Instructions:

Attempt Any Four Questions in all.

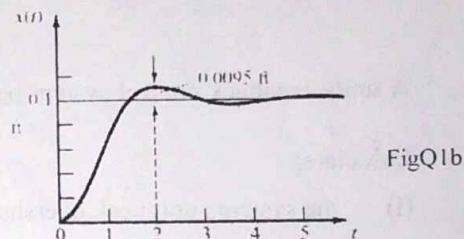
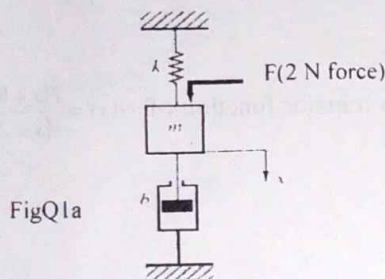
Time Allowed:

2½ HOURS.

Note: State all assumptions made.

Question 1

- a. FigQ1a shows a mechanical vibratory system. When 2 N of force (step input) is applied to the system, the mass oscillates, as shown in FigQ1b. Determine m , b , and k of the system from this response curve. The displacement x is measured from the equilibrium position. **(13 Marks each)**



- b. A unity feedback (negative) system has open-loop transfer function $G(s) = \frac{K}{s(s+2)}$.
- Calculate the value of gain K so that the closed-loop system has a steady state unit ramp error of 0.1. **(3 Marks)**
 - What are the corresponding damping factor and percentage peak overshoot to unit step input. **(3 Marks)**
 - The system is now modified to include a forward path zero at $s = -6$. What is the new value of K for the same steady state error as in part (i). **(3 Marks)**
 - calculate the damping factor and the percentage peak overshoot corresponding to this damping factor. **(3 Marks)**

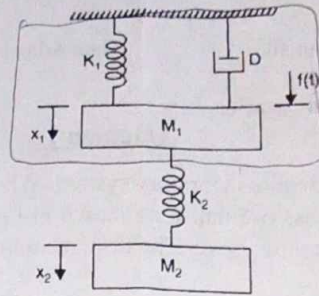
Question 2

- a. The open – loop transfer function of a unity feedback system is $G(s) = \frac{K}{s(s+2)}$. The system is required to meet the following specifications. Peak time, $t_p = 1.0$ sec, Peak overshoot, $M_p = 10\%$.
- Determine the value of K to meet both these specifications? If not, what should be the value of K so that t_p and M_p are violated by the same percentage. **(4 Marks)**
 - Assume that the open-loop poles $s = -2$ can be adjusted. Let it be $s = -\alpha$. Determine the values of K and α in order to meet both the specifications. **(7 Marks)**
 - Suggest a suitable method of adjusting α and draw the block diagram of the modified system. **(4 Marks)**

b. (i) Derive the transfer function $\frac{x_1(s)}{F(s)}$ for the mechanical system shown in FigQ2b

(6 Marks)

(ii) Draw its electrical analog circuit in Force – voltage method and write out the circuit equations. (4 Marks)



FigQ2b

Question 3

a. A unity feedback control system has an open-loop transfer function of $G(s) = \frac{2(s+8)}{s(s+2)}$.

Calculate:

(i) the systems unit peak overshoot. (3 Marks)

(ii) $c(t)$ for unit step input. (5 Marks)

(i) Steady state error, e_{ss} for unit ramp input. (2 Marks)

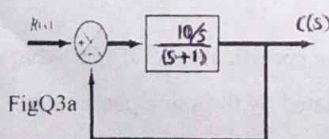
(ii) remove the zero $s = -8$ and increase the gain to 16. Find now damping, ρ , and ω_n . (2 Marks) (v) Calculate $c(t)$ for unit step ramp input. (3 Marks)

b. Consider the system shown in FigQ3a. The damping ratio of this system is 0.158 and the undamped natural frequency is 3.16 rad/sec. To improve the relative stability, a tachometer feedback is employed. FigQ3b shows such a tachometer-feedback system.

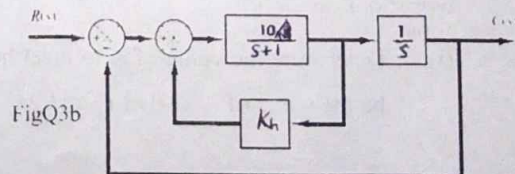
(i) Determine the value of K_h so that the damping ratio of the system is 0.5. (5 Marks)

(ii) Draw unit-step response curves of both the original and tachometer-feedback systems. (3 Marks)

(iii) draw the error-versus-time curves for the unit-ramp response of both systems. (2 Marks)



FigQ3a



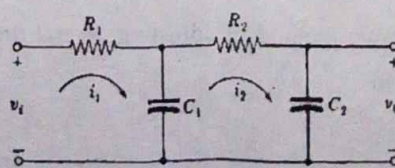
FigQ3b

Question 4

a. (i) Draw the block diagram of the two simple lag networks connected in series in

FigQ4a

(6 Marks)

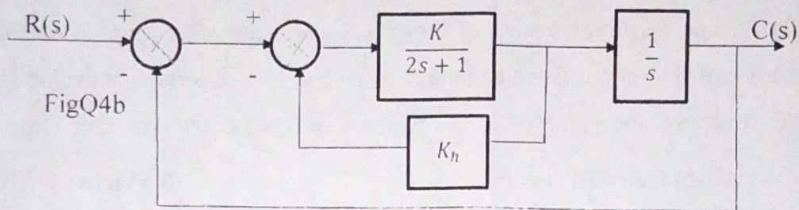


FigQ4a

- (iii) Obtain the transfer function of the network from the block diagram drawn in a(i) (5 Marks)

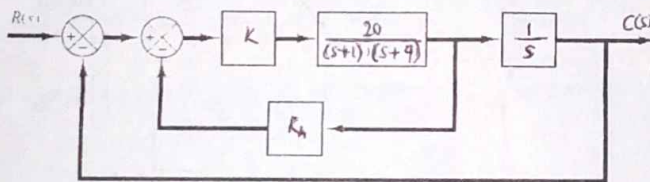
b. Consider the system shown in FigQ4b, which involves velocity feedback.

- (i) Determine the values of the amplifier gain K ; and (7 Marks)
 (ii) the velocity feedback gain K_h so that the following specifications are satisfied:
 Damping ratio of the closed loop poles is 0.5, settling time ≤ 2 sec, Static velocity error constant $K_v \geq 50 \text{ sec}^{-1}$ and $0 < K_h < 1$ (7 Marks)



Question 5

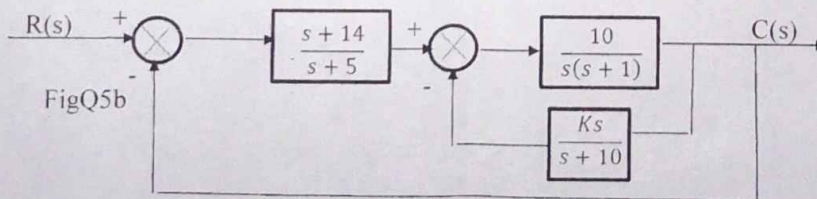
- a. Consider the servo system with tachometer feedback shown in FigQ5a. Determine the ranges of stability for K and K_h . (Note that K_h must be positive.) (12 Marks)



FigQ5a

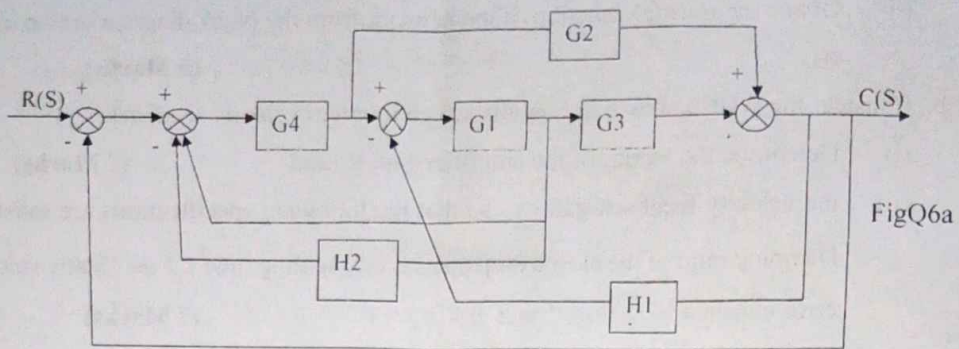
- b. Consider the system in FigQ5b. (i) At what value of K will the system sustain oscillation? (9 Marks)

- (iii) Find the static velocity error constant of the system with this value of K . (4 Marks)



Question 6

- A Mechanical system is represented by the block diagram shown in Fig Q6a. Draw the signal flow graph of the system and determine the transfer function of the system using Manson's gain formula. (12 Marks)

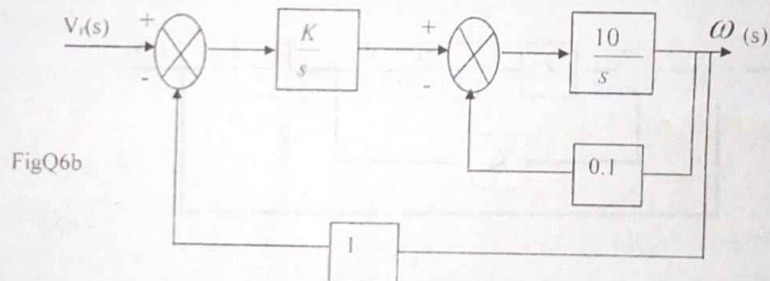


FigQ6a

- b. Consider the speed control system of FigQ6b wherein the inner loop corresponds to motor back emf. The controller is an integrator with gain K observe that the load inertia only. (i) determine the value of K for which steady-state error to unit ramp input ($V_r(s) = \frac{1}{s^2}$) is less than 0.01 rad/sec. (5 Marks)

- (ii) For the value of K obtained in (i), determine the sensitivity $S'_K \cdot T(s) = \frac{\omega(s)}{V_r(s)}$; (5 Marks)

- (iii) what will be the limiting value of S'_K at low frequencies. (5 Marks)



FigQ6b